Software Requirement Specification (SRS)

CareSync - Hospital Workflow Optimization

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1. Introduction:

1.1Purpose

This Software Requirements Specification (SRS) document describes the functional and non functional requirements of the Smart Hospital Workflow & Appointment Optimization System (SHWAS).

The system aims to improve hospital efficiency by optimizing patient flow, reducing waiting time, managing doctor schedules, and prioritizing emergency cases through intelligent decision support.

1.2Scope

The Smart Hospital Workflow & Appointment Optimization System is a web-based application designed to digitize and optimize hospital operations.

The system allows:

- Patients to book and manage appointments online
- Doctors to manage schedules and workload efficiently
- Hospital administrators to monitor analytics and optimize resources
- Emergency cases to be prioritized automatically
- This system functions as a Decision Support System (DSS) for hospital management.

1.3Definitions, Acronyms, and Abbreviations

Term	Description
DSS	Decision Support System
SHWAS	Smart Hospital Workflow & Appointment System

Term	Description
UI	User Interface
API	Application Programming Interface
RBAC	Role-Based Access Control
OTP	One-Time Password

1.4 References

* IEEE Std 830–1998: IEEE Recommended Practice for Software Requirements Specifications
* Hospital workflow management case studies
* Government hospital operational guidelines.

1.5 Document Overview

This document contains:

1. Overall system description
2. Functional and non-functional requirements
3. Interface requirements
4. System constraints and assumptions

2. Overall Description:

2.1 Product Perspective

SHWAS is a web-based multi-module system consisting of:

1.Patient Dashboard

2. Doctor Dashboard

3. Admin / Management Dashboard

4. Each module operates under role-based authentication and provides specific functionality.

2.2 Product Functions

- Online appointment booking and rescheduling
- Real-time waiting time prediction
- Emergency case prioritization
- Doctor workload balancing
- Administrative analytics and reporting

2.3 User Characteristics

User Type	Description
Patient	ointments and views waiting time Manages
Doctor	ity and workload Administrator Monitors
	ospital performance and resources

2.4 Operating Environment

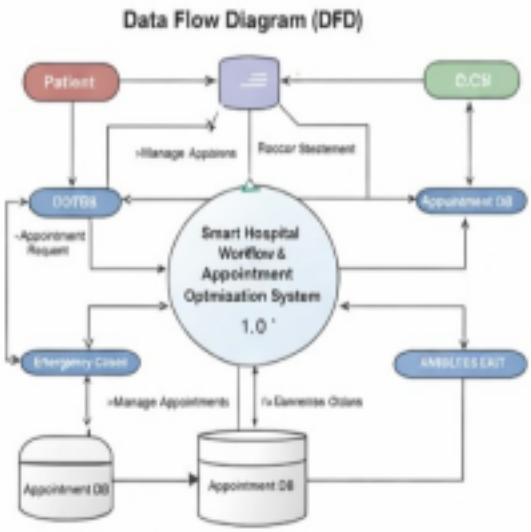
- **Frontend**: React / Angular with HTML, CSS, Bootstrap
- **Backend:** Spring Boot / Django / Node.js
- **Database**: MySQL / PostgreSQL / MongoDB
- **Analytics**: Python (Pandas, NumPy)
- **Web browsers**: Chrome, Edge, Firefox

2.5 Design Constraints

- Role-based authentication is mandatory
- Emergency workflows must execute with minimal delay
- Patient data must be securely stored and transmitted
- System must support concurrent users

2.6 Assumptions and Dependencies

- Doctors are registered by hospital administration
- Patients provide accurate information
- Analytics accuracy depends on historical data
- Internet connectivity is available



3. Specific Requirements:

3.1 External Interfaces

3.1.1 User Interfaces

• The system shall provide separate web interfaces for Pa7ents, Doctors, and Administrators. • Pa7ent interface includes appointment booking, wai7ng 7me view, and emergency flagging • Doctor interface includes schedule management and workload view • Admin interface includes analy7cs dashboards and reports

3.1.2 Communication Interfaces

• The system shall use HTTP/HTTPS protocols

• Secure APIs shall be used for data exchange

3.2 Functions

3.2.1 Public User (Citizen) Functions

3.2.1.1 Registration and Login

• The system shall allow patients to register and log in securely • The system shall support OTP-based authentication

3.2.1.2 Appointment Management

• The system shall allow pa7ents to book appointments • The system shall predict approximate wai7ng 7me • The system shall allow appointment rescheduling

3.2.1.3 Emergency Handling

• The system shall allow pa7ents to flag emergency cases • The system shall priori7ze emergency pa7ents automa7cally

3.2.2 Doctor Functions

3.2.2.1 Doctor Authentication

• The system shall allow doctors to log in securely. • The system shall provide role-based access.

3.2.2.2 Schedule Management

• The system shall allow doctors to manage availability •
The system shall display daily workload

3.2.2.3 Performance Analytics

• The system shall provide performance and workload analy7cs.

3.2.3 Administrator Functions

3.2.3.1 Admin Authentication

• The system shall provide secure admin login

3.2.3.2 Hospital Management

• The system shall manage doctors and departments •
The system shall monitor pa7ent flow and peak hours

3.2.3.3 Reports

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• The system shall generate u7liza7on and performance reports

3.3 Performance Requirements

• The system shall support at least 100 concurrent users •

Response 7me shall be less than 2 seconds under normal load

3.4 Logical Database Requirements

The system shall store:

- Patient data
- Doctor data
- Appointment details
- Emergency case records

- Administrative reports

3.5 Design Constraints

- Web-based architecture.
- Secure role-based access control.

3.6 Software System Quality Attributes

3.6.1 Security

- Encrypted authen7ca7on

- Secure data storage
- Role-based authoriza7on

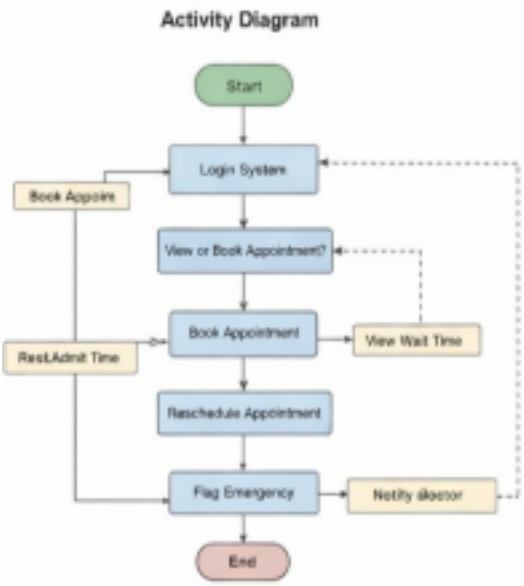
3.6.2 Reliability and Availability

- High availability for cri7cal hospital opera7ons.

3.7 Object-Oriented Models

The system shall include:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- ER Diagram



4.External Interface Requiremetss:

4.1 User Interface

- Separate dashboards for Patient, Doctor, and Admin

4.2 Hardware Interfaces

- Server-hosted application

4.3 Software Interfaces

- Database server
- Analytics libraries

4.4 Communication Interfaces

- HTTP/HTTPS protocols

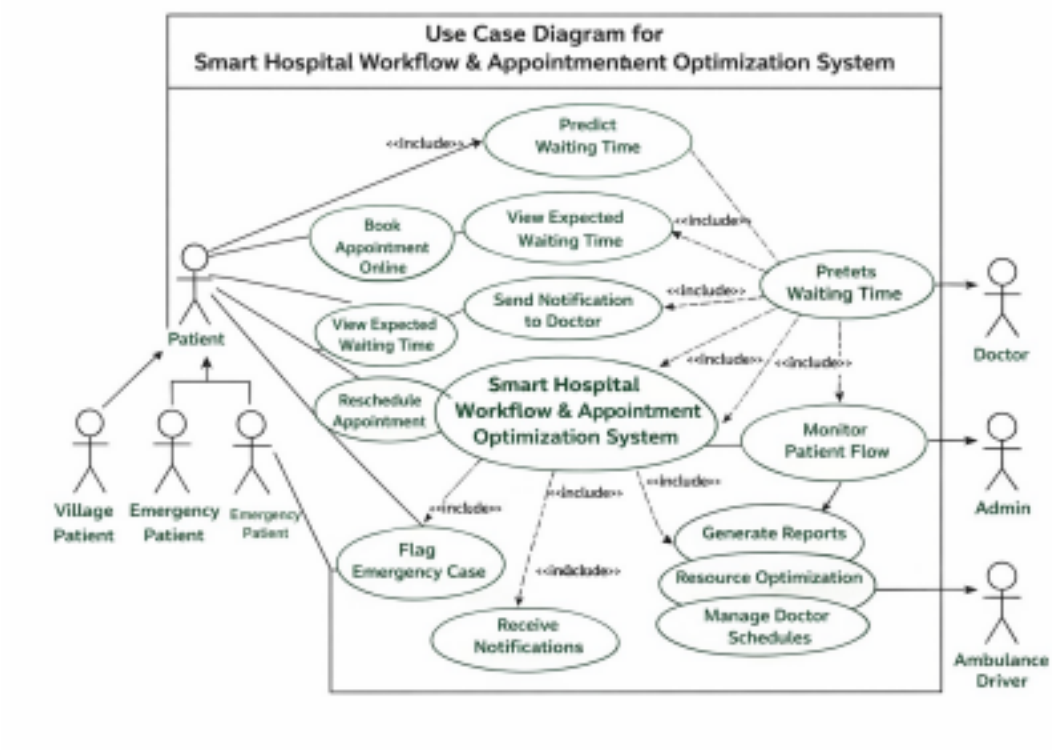
5.Future Enhancements

- AI-based disease prediction

● IoT-enabled smart hospital infrastructure

● Mobile application

● Pharmacy and lab integration



6.Conclusion:

The Smart Hospital Workflow & Appointment Optimization System is designed to address critical challenges faced by modern hospitals, such as long patient waiting times, inefficient appointment scheduling, and poor coordination among departments. By integrating intelligent scheduling, real-time workflow monitoring, and automated notifications, the system enhances operational efficiency and improves the overall patient experience.

In conclusion, the Smart Hospital Workflow & Appointment Optimization System has the potential to significantly modernize hospital management processes, reduce administrative burden, and support high-quality healthcare delivery. With future enhancements such as AI-based predictions and telemedicine integration, the system can evolve into a comprehensive smart healthcare solution.